



Sri Eshwar
EDUCATION

GOWTHAM ARUL I

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| **Linkedin:** [gowtham-arul](#) | **GitHub:** [Gowtham-Arul](#) | **Portfolio**

Sri Eshwar College of Engineering	B.E ECE	8.85 (till III Sem)	2024-2028
Velammal International School (CBSE)	HSC	88%	2022-2024
Pearls Public School (CBSE)	SSLC	97.6%	2021-2022

INTERNSHIPS

IIIT SriCity **May-Jun 2026**
Completed a one month summer internship where I worked with pre-trained **CNN model (ResNet-50)**, including adding image augmentation to improve recognition accuracy. Got exposure to **ESP32-CAM MB** module for capturing and processing image data. Worked with **Raspberry Pi 3** by integrating sensors to it and gained hands-on experience with Raspbian OS, understanding how a single-board computer operates in real-time tasks.

NIELIT Calicut (Embedded Systems & IoT) **Dec 2025** [Link](#)
Completed a one-month Winter internship focused on Embedded Systems and IoT basics. Gained practical experience in writing Embedded C programs and simulating sensors and actuators using **Wokwi**. Understood various **communication protocols** and got hands-on exposure to cloud platforms like **Adafruit IO** for real-time data monitoring.

PROJECTS

AI-Enabled Facial Recognition Attendance System **2026** [GitHub](#)
Built an end-to-end facial recognition based attendance system that **streams live video** from a camera module, identifies faces using a **pre-trained CNN model**, and automatically marks attendance in real time.
Tech Stack: ESP32-CAM MB Module, Python, ResNet-50 (CNN) + Data Augmentation, Scikit-learn, Node.js, React.js, Tailwind CSS, MySQL, MJPEG over HTTP
Skills Acquired: Face recognition using pre-trained CNN, anti-spoofing techniques and full-stack integration from hardware to web interface.

SmartShield - IoT Enabled Helmet for Mining Safety **2025** [GitHub](#)
Built a **smart safety helmet** that continuously monitors toxic gases, temperature, humidity, and worker heart rate, showing real-time alerts to **prevent accidents** in hazardous environments.
Tech Stack: ESP8266, Gas, Temperature and Humidity Sensor, Pulse Sensor, OLED Display, Embedded C++, Arduino IDE, HTTP Protocol
Skills Acquired: Sensor interfacing and data acquisition, hardware-software integration on ESP8266, I2C protocol for OLED display communication, and real-time data monitoring over HTTP.

CERTIFICATIONS

Design Thinking - A primer	NPTEL	2026
Digital Circuits	NPTEL	2025
Complete Guide to C Programming	LinkedIn Learning	2025
Mastering Data Structures & Algorithms using C & C++	Udemy	2025
C Training	IIT Bombay	2024

ACHIEVEMENTS

Won 2 nd Prize in Mini Project Expo held for 2 nd year students	2026
Participated in TechnoMind 2K25 (Project Expo in Karpagam Institute of Technology)	2025
Participated in Freshathon (Internal project Expo)	2025
Selected as Finalist in selfE Hackathon (Internal Hackathon)	2024
Skill Rack Solved 1000+ Problems 308 Bronzes 6 Certificates Link	

SKILLS

Languages	C Embedded C Python (Basic)
Core Concepts	Circuit basics Semiconductor Devices Digital Electronics Microcontrollers
Hardware	8085 8051 ESP32 ESP8266 Raspberry pi 3 Sensors and Actuators Interfacing
Software	Web Development : HTML, CSS, JavaScript React Framework
Database	MySQL
IDE	Arduino IDE VS Code Eclipse
Tools	Keil Proteus Wokwi Proto Cloud Platforms : Adafruit IO, ThingSpeak Multisim Octave Canva Google COLAB